

Compiled experiment results

Statlog German Credit and Default of Credit Card Client

This report gathers live classifier scores, cross-validation where it exists, Algorithm 2 permutation tests, sampling-ratio audits, and revised-snapshot protocol scores for the two paper datasets. It covers the tabular ML baselines and the eight named TDA processes (split $\times$ undersample $\times$ just H0 vs both H0 and H1). Public names always use “and”, never “+”. Nested extras (drop-correlated columns, linear regression, mean/variance) are archived and are not included.

Generated 2026-08-27 from artefacts under 6_Results/. H0 processes slice homology-0 barcode statistics from the matching H0-and-H1 run; they do not rerun Ripser. Algorithm 2 on the four just-H0 folders has not been generated yet. Green rows mark the highest F1 in a table when that comparison is meaningful. Cross-validation is training-set resampling and is not mixed with held-out test scores.

Coverage

Dataset	Process / family	Experiment	Artefacts
Statlog German Credit	Tabular baseline	1 ML Default Parameters	Test metrics and CV
Statlog German Credit	Tabular baseline	2 ML Tuned Parameters	Test metrics and CV
Statlog German Credit	Early split and undersample, using just H0	Live process	Default classifiers; Algorithm 2 not generated
Statlog German Credit	Early split and undersample, using both H0 and H1	Live process	Default classifiers; tuned; Algorithm 2; revised protocol
Statlog German Credit	Early split, no undersample, using just H0	Live process	Default classifiers; Algorithm 2 not generated
Statlog German Credit	Early split, no undersample, using both H0 and H1	Live process	Default classifiers; tuned; Algorithm 2; revised protocol
Statlog German Credit	Late split and undersample (the original historical run), using just H0	Live process	Default classifiers; Algorithm 2 not generated
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Statlog German Credit	Late split, no undersample, using just H0	Live process	Default classifiers; Algorithm 2 not generated
Statlog German Credit	Late split, no undersample, using both H0 and H1	Live process	Default classifiers; tuned; Algorithm 2; revised protocol
Default of Credit Card Client	Tabular baseline	1 ML Default Parameters	Test metrics and CV
Default of Credit Card Client	Tabular baseline	2 ML Tuned Parameters	Test metrics and CV
Default of Credit Card Client	Early split and undersample, using just H0	Live process	Default classifiers; Algorithm 2 not generated
Default of Credit Card Client	Early split and undersample, using both H0 and H1	Live process	Default classifiers; tuned; Algorithm 2; revised protocol
Default of Credit Card Client	Early split, no undersample, using just H0	Live process	Default classifiers; Algorithm 2 not generated

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Statlog German Credit

Headline numbers below take, for each process and snapshot size, the classifier with the highest held-out F1. Full model-by-model tables follow.

Headline held-out F1

Process	Snapshot / setting	Best model	Accuracy	Precision	Recall	F1	Status
Tabular ML, default parameters	Original features	Logistic regression	0.745	0.565	0.650	0.605	Available
Tabular ML, tuned parameters	Original features	Logistic regression	0.710	0.514	0.633	0.567	Available
Early split and undersample, using just H0	L30	KNN	0.510	0.505	1.000	0.671	Available
Early split and undersample, using just H0	L60	KNN	0.500	0.500	1.000	0.667	Available
Early split and undersample, using both H0 and H1	L30	KNN	0.508	0.504	0.998	0.670	Available
Early split and undersample, using both H0 and H1	L60	KNN	0.500	0.500	1.000	0.667	Available
Early split, no undersample, using just H0	L30	SVM	0.500	0.500	1.000	0.667	Available
Early split, no undersample, using just H0	L60	SVM	0.500	0.500	1.000	0.667	Available
Early split, no undersample, using both H0 and H1	L30	SVM	0.500	0.500	1.000	0.667	Available
Early split, no undersample, using both H0 and H1	L60	SVM	0.500	0.500	1.000	0.667	Available
Late split and undersample (the original historical run), using just H0	L30	Random forest	0.730	0.689	0.840	0.757	Available
Late split and undersample (the original historical run), using just H0	L60	XGBoost	0.860	0.827	0.910	0.867	Available
Late split and undersample (the original historical run), using both H0 and H1	L30	Random forest	0.765	0.739	0.820	0.777	Available
Late split and undersample (the original historical run), using both H0 and H1	L60	XGBoost	0.850	0.837	0.870	0.853	Available
Late split, no undersample, using just H0	L30	KNN	1.000	1.000	1.000	1.000	Available
Late split, no undersample, using just H0	L60	SVM	1.000	1.000	1.000	1.000	Available
Late split, no undersample, using both H0 and H1	L30	SVM	1.000	1.000	1.000	1.000	Available
Late split, no undersample, using both H0 and H1	L60	SVM	1.000	1.000	1.000	1.000	Available

Tabular ML baselines

Tabular features, default hyperparameters

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
Original features	SVM	0.740	0.559	0.633	0.594
Original features	KNN	0.680	0.475	0.633	0.543
Original features	XGBoost	0.755	0.622	0.467	0.533
Original features	Logistic regression	0.745	0.565	0.650	0.605
Original features	Random forest	0.750	0.609	0.467	0.528

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
5-fold CV	SVM	0.762	0.026
5-fold CV	KNN	0.724	0.028
5-fold CV	XGBoost	0.741	0.041
5-fold CV	Logistic regression	0.767	0.027
5-fold CV	Random forest	0.753	0.016

Tabular features, GridSearchCV

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
Original features	SVM	0.730	0.554	0.517	0.534
Original features	KNN	0.710	0.517	0.500	0.508
Original features	XGBoost	0.735	0.566	0.500	0.531
Original features	Logistic regression	0.710	0.514	0.633	0.567
Original features	Random forest	0.750	0.593	0.533	0.561

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
5-fold CV	SVM	0.730	0.046
5-fold CV	KNN	0.735	0.029
5-fold CV	XGBoost	0.779	0.024
5-fold CV	Logistic regression	0.776	0.034
5-fold CV	Random forest	0.769	0.015

Early split and undersample, using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.421	0.409	0.354	0.379
L30	KNN	0.510	0.505	1.000	0.671
L30	XGBoost	0.505	0.503	0.994	0.668
L30	Logistic regression	0.503	0.502	1.000	0.668
L30	Random forest	0.508	0.504	0.990	0.668
L60	SVM	0.372	0.198	0.084	0.118
L60	KNN	0.500	0.500	1.000	0.667
L60	XGBoost	0.500	0.500	1.000	0.667
L60	Logistic regression	0.500	0.500	1.000	0.667
L60	Random forest	0.500	0.500	1.000	0.667

Algorithm 2 has not been generated for this process on this dataset.

Early split and undersample, using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.436	0.275	0.078	0.121
L30	KNN	0.508	0.504	0.998	0.670
L30	XGBoost	0.504	0.502	0.984	0.665
L30	Logistic regression	0.503	0.502	1.000	0.668
L30	Random forest	0.504	0.502	0.978	0.664
L60	SVM	0.493	0.294	0.010	0.019
L60	KNN	0.500	0.500	1.000	0.667
L60	XGBoost	0.500	0.500	1.000	0.667
L60	Logistic regression	0.500	0.500	1.000	0.667
L60	Random forest	0.500	0.500	1.000	0.667

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.503	0.502	1.000	0.668
L30	KNN	0.506	0.503	0.996	0.668
L30	XGBoost	0.498	0.499	0.976	0.660
L30	Logistic regression	0.502	0.501	1.000	0.668
L30	Random forest	0.506	0.503	0.994	0.668
L60	SVM	0.500	0.500	1.000	0.667
L60	KNN	0.500	0.500	1.000	0.667
L60	XGBoost	0.500	0.500	1.000	0.667
L60	Logistic regression	0.500	0.500	1.000	0.667
L60	Random forest	0.500	0.500	1.000	0.667

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small p -values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L30	(2, 2)	0.1336	0.005	100	100
L30	(1, 1)	1.1377	0.005	100	100
L30	(2, 1)	0.3425	0.005	100	100
L30	(2, 2)	0.4187	0.065	100	100
L30	(1, 1)	2.2016	0.110	100	100
L30	(2, 1)	0.6082	0.075	100	100
L60	(2, 2)	0.0614	0.005	100	100
L60	(1, 1)	0.7109	0.005	100	100
L60	(2, 1)	0.2261	0.005	100	100
L60	(2, 2)	0.1417	0.005	100	100
L60	(1, 1)	1.2415	0.005	100	100
L60	(2, 1)	0.3556	0.005	100	100

Sampling-ratio audit ($H0$ and $H1$ processes only). Reuse ratio = (points per snapshot $\times$ number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
30.0	historical l500	72	500	150.00	class1
30.0	revised ceil n over t	72	4	1.20	class1
30.0	historical l500	72	500	150.00	class2

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
30.0	revised ceil n over t	72	4	1.20	class2
60.0	historical l500	144	500	300.00	class1
60.0	revised ceil n over t	144	2	1.20	class1
60.0	historical l500	144	500	300.00	class2
60.0	revised ceil n over t	144	2	1.20	class2
30.0	historical l500	18	500	150.00	class1
30.0	revised ceil n over t	18	4	1.20	class1
30.0	historical l500	18	500	150.00	class2
30.0	revised ceil n over t	18	4	1.20	class2
60.0	historical l500	36	500	300.00	class1
60.0	revised ceil n over t	36	2	1.20	class1
60.0	historical l500	36	500	300.00	class2
60.0	revised ceil n over t	36	2	1.20	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). H0 and H1 processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.458	0.458	0.462	0.500	0.480	0.514
Random forest	0.625	0.625	0.667	0.500	0.571	0.743
XGBoost	0.667	0.667	0.750	0.500	0.600	0.663
SVM	0.500	0.500	0.500	0.500	0.500	0.556
KNN	0.708	0.708	0.727	0.667	0.696	0.681
Logistic regression	0.500	0.500	0.500	0.500	0.500	0.493
Random forest	0.458	0.458	0.467	0.583	0.519	0.618
XGBoost	0.542	0.542	0.538	0.583	0.560	0.528
SVM	0.583	0.583	0.556	0.833	0.667	0.514
KNN	0.542	0.542	0.545	0.500	0.522	0.528

Early split, no undersample, using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.500	0.500	1.000	0.667
L30	KNN	0.500	0.500	1.000	0.667
L30	XGBoost	0.500	0.500	1.000	0.667
L30	Logistic regression	0.500	0.500	1.000	0.667
L30	Random forest	0.500	0.500	1.000	0.667
L60	SVM	0.500	0.500	1.000	0.667
L60	KNN	0.500	0.500	1.000	0.667
L60	XGBoost	0.500	0.500	1.000	0.667
L60	Logistic regression	0.500	0.500	1.000	0.667
L60	Random forest	0.500	0.500	1.000	0.667

Algorithm 2 has not been generated for this process on this dataset.

Early split, no undersample, using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.500	0.500	1.000	0.667
L30	KNN	0.500	0.500	1.000	0.667
L30	XGBoost	0.500	0.500	1.000	0.667
L30	Logistic regression	0.500	0.500	1.000	0.667
L30	Random forest	0.500	0.500	1.000	0.667
L60	SVM	0.500	0.500	1.000	0.667
L60	KNN	0.500	0.500	1.000	0.667
L60	XGBoost	0.500	0.500	1.000	0.667
L60	Logistic regression	0.500	0.500	1.000	0.667
L60	Random forest	0.500	0.500	1.000	0.667

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.500	0.500	1.000	0.667

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	KNN	0.500	0.500	1.000	0.667
L30	XGBoost	0.500	0.500	1.000	0.667
L30	Logistic regression	0.500	0.500	1.000	0.667
L30	Random forest	0.500	0.500	1.000	0.667
L60	SVM	0.500	0.500	1.000	0.667
L60	KNN	0.500	0.500	1.000	0.667
L60	XGBoost	0.500	0.500	1.000	0.667
L60	Logistic regression	0.500	0.500	1.000	0.667
L60	Random forest	0.500	0.500	1.000	0.667

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small p -values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L30	(2, 2)	0.1105	0.005	100	100
L30	(1, 1)	0.9934	0.005	100	100
L30	(2, 1)	0.3102	0.005	100	100
L30	(2, 2)	0.3374	0.005	100	100
L30	(1, 1)	1.9286	0.005	100	100
L30	(2, 1)	0.5441	0.005	100	100
L60	(2, 2)	0.0488	0.005	100	100
L60	(1, 1)	0.5973	0.005	100	100
L60	(2, 1)	0.1997	0.005	100	100
L60	(2, 2)	0.1255	0.005	100	100
L60	(1, 1)	1.0961	0.005	100	100
L60	(2, 1)	0.3280	0.005	100	100

Sampling-ratio audit (H0 and H1 processes only). Reuse ratio = (points per snapshot $\times$ number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
30.0	historical l500	72	500	150.00	class1
30.0	revised ceil n over t	72	4	1.20	class1
30.0	historical l500	168	500	150.00	class2
30.0	revised ceil n over t	168	4	1.20	class2

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
60.0	historical l500	144	500	300.00	class1
60.0	revised ceil n over t	144	2	1.20	class1
60.0	historical l500	336	500	300.00	class2
60.0	revised ceil n over t	336	2	1.20	class2
30.0	historical l500	18	500	150.00	class1
30.0	revised ceil n over t	18	4	1.20	class1
30.0	historical l500	42	500	150.00	class2
30.0	revised ceil n over t	42	4	1.20	class2
60.0	historical l500	36	500	300.00	class1
60.0	revised ceil n over t	36	2	1.20	class1
60.0	historical l500	84	500	300.00	class2
60.0	revised ceil n over t	84	2	1.20	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). H0 and H1 processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.667	0.667	0.625	0.833	0.714	0.750
Random forest	0.583	0.583	0.571	0.667	0.615	0.722
XGBoost	0.583	0.583	0.571	0.667	0.615	0.569
SVM	0.583	0.583	0.556	0.833	0.667	0.194
KNN	0.417	0.417	0.429	0.500	0.462	0.486
Logistic regression	0.500	0.500	0.500	0.500	0.500	0.444
Random forest	0.583	0.583	0.600	0.500	0.545	0.417
XGBoost	0.583	0.583	0.600	0.500	0.545	0.444
SVM	0.500	0.500	0.500	0.500	0.500	0.500
KNN	0.500	0.500	0.500	0.500	0.500	0.458
Logistic regression	0.417	0.417	0.444	0.667	0.533	0.500
Random forest	0.500	0.500	0.500	0.500	0.500	0.500
XGBoost	0.583	0.583	0.600	0.500	0.545	0.472
SVM	0.417	0.417	0.444	0.667	0.533	0.417
KNN	0.500	0.500	0.500	0.333	0.400	0.458

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.542	0.542	0.556	0.417	0.476	0.472
Random forest	0.458	0.458	0.429	0.250	0.316	0.417
XGBoost	0.458	0.458	0.429	0.250	0.316	0.385
SVM	0.417	0.417	0.400	0.333	0.364	0.562
KNN	0.500	0.500	0.500	0.333	0.400	0.524
Logistic regression	0.708	0.708	0.692	0.750	0.720	0.743
Random forest	0.542	0.542	0.556	0.417	0.476	0.451
XGBoost	0.417	0.417	0.417	0.417	0.417	0.340
SVM	0.625	0.625	0.588	0.833	0.690	0.257
KNN	0.500	0.500	0.500	0.417	0.455	0.531

Late split and undersample (the original historical run), using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.690	0.644	0.850	0.733
L30	KNN	0.640	0.632	0.670	0.650
L30	XGBoost	0.700	0.682	0.750	0.714
L30	Logistic regression	0.670	0.649	0.740	0.692
L30	Random forest	0.730	0.689	0.840	0.757
L60	SVM	0.735	0.701	0.820	0.756
L60	KNN	0.775	0.739	0.850	0.791
L60	XGBoost	0.860	0.827	0.910	0.867
L60	Logistic regression	0.720	0.700	0.770	0.733
L60	Random forest	0.820	0.791	0.870	0.829

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L30	SVM	0.671	0.036
L30	KNN	0.685	0.031

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L30	XGBoost	0.680	0.040
L30	Logistic regression	0.671	0.029
L30	Random forest	0.698	0.017
L60	SVM	0.690	0.061
L60	KNN	0.794	0.032
L60	XGBoost	0.858	0.040
L60	Logistic regression	0.684	0.055
L60	Random forest	0.809	0.049

Algorithm 2 has not been generated for this process on this dataset.

Late split and undersample (the original historical run), using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.740	0.697	0.850	0.766
L30	KNN	0.685	0.646	0.820	0.722
L30	XGBoost	0.745	0.713	0.820	0.763
L30	Logistic regression	0.725	0.706	0.770	0.737
L30	Random forest	0.765	0.739	0.820	0.777
L60	SVM	0.825	0.793	0.880	0.834
L60	KNN	0.775	0.743	0.840	0.789
L60	XGBoost	0.850	0.837	0.870	0.853
L60	Logistic regression	0.805	0.770	0.870	0.817
L60	Random forest	0.800	0.788	0.820	0.804

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L30	SVM	0.706	0.045
L30	KNN	0.672	0.045
L30	XGBoost	0.721	0.029

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L30	Logistic regression	0.711	0.048
L30	Random forest	0.731	0.030
L60	SVM	0.810	0.035
L60	KNN	0.829	0.022
L60	XGBoost	0.861	0.028
L60	Logistic regression	0.809	0.030
L60	Random forest	0.859	0.018

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.725	0.696	0.800	0.744
L30	KNN	0.735	0.697	0.830	0.758
L30	XGBoost	0.730	0.698	0.810	0.750
L30	Logistic regression	0.720	0.696	0.780	0.736
L30	Random forest	0.740	0.707	0.820	0.759
L60	SVM	0.825	0.793	0.880	0.834
L60	KNN	0.775	0.743	0.840	0.789
L60	XGBoost	0.840	0.804	0.900	0.849
L60	Logistic regression	0.790	0.764	0.840	0.800
L60	Random forest	0.810	0.798	0.830	0.814

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L30	SVM	0.725	0.045
L30	KNN	0.700	0.043
L30	XGBoost	0.729	0.026
L30	Logistic regression	0.724	0.030
L30	Random forest	0.726	0.038
L60	SVM	0.810	0.035
L60	KNN	0.829	0.022
L60	XGBoost	0.870	0.018

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L60	Logistic regression	0.831	0.023
L60	Random forest	0.860	0.018

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small *p*-values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L30	(2, 2)	0.1015	0.005	100	100
L30	(1, 1)	1.0151	0.005	100	100
L30	(2, 1)	0.2996	0.005	100	100
L60	(2, 2)	0.0460	0.005	100	100
L60	(1, 1)	0.6368	0.005	100	100
L60	(2, 1)	0.1988	0.005	100	100

Sampling-ratio audit (*H0* and *H1* processes only). Reuse ratio = (points per snapshot × number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
30.0	historical l500	90	500	150.00	class1
30.0	revised ceil n over t	90	4	1.20	class1
30.0	historical l500	90	500	150.00	class2
30.0	revised ceil n over t	90	4	1.20	class2
60.0	historical l500	180	500	300.00	class1
60.0	revised ceil n over t	180	2	1.20	class1
60.0	historical l500	180	500	300.00	class2
60.0	revised ceil n over t	180	2	1.20	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). *H0* and *H1* processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.458	0.458	0.444	0.333	0.381	0.368
Random forest	0.417	0.417	0.429	0.500	0.462	0.392
XGBoost	0.417	0.417	0.429	0.500	0.462	0.306
SVM	0.333	0.333	0.400	0.667	0.500	0.333
KNN	0.375	0.375	0.385	0.417	0.400	0.413
Logistic regression	0.583	0.583	0.667	0.333	0.444	0.458
Random forest	0.458	0.458	0.429	0.250	0.316	0.396

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
XGBoost	0.458	0.458	0.444	0.333	0.381	0.597
SVM	0.458	0.458	0.455	0.417	0.435	0.542
KNN	0.542	0.542	0.545	0.500	0.522	0.583

Late split, no undersample, using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	0.995	0.990	1.000	0.995
L30	KNN	1.000	1.000	1.000	1.000
L30	XGBoost	1.000	1.000	1.000	1.000
L30	Logistic regression	1.000	1.000	1.000	1.000
L30	Random forest	1.000	1.000	1.000	1.000
L60	SVM	1.000	1.000	1.000	1.000
L60	KNN	1.000	1.000	1.000	1.000
L60	XGBoost	0.995	1.000	0.990	0.995
L60	Logistic regression	1.000	1.000	1.000	1.000
L60	Random forest	1.000	1.000	1.000	1.000

Algorithm 2 has not been generated for this process on this dataset.

Late split, no undersample, using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	1.000	1.000	1.000	1.000
L30	KNN	1.000	1.000	1.000	1.000
L30	XGBoost	1.000	1.000	1.000	1.000
L30	Logistic regression	1.000	1.000	1.000	1.000
L30	Random forest	1.000	1.000	1.000	1.000
L60	SVM	1.000	1.000	1.000	1.000

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L60	KNN	1.000	1.000	1.000	1.000
L60	XGBoost	0.995	1.000	0.990	0.995
L60	Logistic regression	1.000	1.000	1.000	1.000
L60	Random forest	1.000	1.000	1.000	1.000

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L30	SVM	1.000	1.000	1.000	1.000
L30	KNN	1.000	1.000	1.000	1.000
L30	XGBoost	1.000	1.000	1.000	1.000
L30	Logistic regression	1.000	1.000	1.000	1.000
L30	Random forest	1.000	1.000	1.000	1.000
L60	SVM	1.000	1.000	1.000	1.000
L60	KNN	1.000	1.000	1.000	1.000
L60	XGBoost	0.995	1.000	0.990	0.995
L60	Logistic regression	1.000	1.000	1.000	1.000
L60	Random forest	1.000	1.000	1.000	1.000

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small p-values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L30	(2, 2)	0.0959	0.005	100	100
L30	(1, 1)	0.9171	0.005	100	100
L30	(2, 1)	0.2878	0.005	100	100
L60	(2, 2)	0.0470	0.005	100	100
L60	(1, 1)	0.5834	0.005	100	100
L60	(2, 1)	0.1971	0.005	100	100

Sampling-ratio audit (H0 and H1 processes only). Reuse ratio = (points per snapshot × number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
30.0	historical l500	90	500	150.00	class1
30.0	revised ceil n over t	90	4	1.20	class1
30.0	historical l500	210	500	150.00	class2

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
30.0	revised ceil n over t	210	4	1.20	class2
60.0	historical l500	180	500	300.00	class1
60.0	revised ceil n over t	180	2	1.20	class1
60.0	historical l500	420	500	300.00	class2
60.0	revised ceil n over t	420	2	1.20	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). H0 and H1 processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.500	0.500	0.500	0.417	0.455	0.486
Random forest	0.375	0.375	0.333	0.250	0.286	0.354
XGBoost	0.375	0.375	0.364	0.333	0.348	0.410
SVM	0.458	0.458	0.455	0.417	0.435	0.507
KNN	0.500	0.500	0.500	0.417	0.455	0.507
Logistic regression	0.583	0.583	0.583	0.583	0.583	0.479
Random forest	0.333	0.333	0.333	0.333	0.333	0.392
XGBoost	0.583	0.583	0.583	0.583	0.583	0.576
SVM	0.333	0.333	0.000	0.000	0.000	0.528
KNN	0.458	0.458	0.429	0.250	0.316	0.413

Default of Credit Card Client

Headline numbers below take, for each process and snapshot size, the classifier with the highest held-out F1. Full model-by-model tables follow.

Headline held-out F1

Process	Snapshot / setting	Best model	Accuracy	Precision	Recall	F1	Status
Tabular ML, default parameters	Original features	SVM	0.811	0.595	0.454	0.515	Available
Tabular ML, tuned parameters	Original features	SVM	0.809	0.589	0.455	0.513	Available
Early split and undersample, using just H0	L15	KNN	0.500	0.500	1.000	0.667	Available
Early split and undersample, using just H0	L5	KNN	0.500	0.500	1.000	0.667	Available
Early split and undersample, using both H0 and H1	L15	KNN	0.500	0.500	1.000	0.667	Available
Early split and undersample, using both H0 and H1	L5	KNN	0.500	0.500	1.000	0.667	Available
Early split, no undersample, using just H0	L15	XGBoost	0.911	0.849	1.000	0.918	Available
Early split, no undersample, using just H0	L5	XGBoost	0.553	0.528	1.000	0.691	Available
Early split, no undersample, using both H0 and H1	L15	XGBoost	0.911	0.849	1.000	0.918	Available
Early split, no undersample, using both H0 and H1	L5	XGBoost	0.553	0.528	1.000	0.691	Available
Late split and undersample (the original historical run), using just H0	L15	SVM	1.000	1.000	1.000	1.000	Available
Late split and undersample (the original historical run), using just H0	L5	Random forest	0.990	1.000	0.980	0.990	Available
Late split and undersample (the original historical run), using both H0 and H1	L15	SVM	1.000	1.000	1.000	1.000	Available
Late split and undersample (the original historical run), using both H0 and H1	L5	KNN	0.990	1.000	0.980	0.990	Available
Late split, no undersample, using just H0	L15	SVM	1.000	1.000	1.000	1.000	Available
Late split, no undersample, using just H0	L5	SVM	1.000	1.000	1.000	1.000	Available
Late split, no undersample, using both H0 and H1	L15	SVM	1.000	1.000	1.000	1.000	Available
Late split, no undersample, using both H0 and H1	L5	SVM	1.000	1.000	1.000	1.000	Available

Tabular ML baselines

Tabular features, default hyperparameters

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
Original features	SVM	0.811	0.595	0.454	0.515
Original features	KNN	0.749	0.437	0.475	0.455
Original features	XGBoost	0.813	0.634	0.370	0.468
Original features	Logistic regression	0.678	0.369	0.638	0.467
Original features	Random forest	0.812	0.618	0.398	0.484

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
5-fold CV	SVM	0.817	0.005
5-fold CV	KNN	0.792	0.010
5-fold CV	XGBoost	0.813	0.006
5-fold CV	Logistic regression	0.810	0.004
5-fold CV	Random forest	0.814	0.008

Tabular features, GridSearchCV

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
Original features	SVM	0.809	0.589	0.455	0.513
Original features	KNN	0.750	0.430	0.403	0.416
Original features	XGBoost	0.816	0.646	0.372	0.472
Original features	Logistic regression	0.678	0.369	0.638	0.467
Original features	Random forest	0.813	0.622	0.394	0.483

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
5-fold CV	SVM	0.820	0.007
5-fold CV	KNN	0.765	0.010
5-fold CV	XGBoost	0.816	0.007
5-fold CV	Logistic regression	0.810	0.004
5-fold CV	Random forest	0.816	0.007

Early split and undersample, using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	0.500	0.000	0.000	0.000
L15	KNN	0.500	0.500	1.000	0.667
L15	XGBoost	0.500	0.500	1.000	0.667
L15	Logistic regression	0.500	0.500	1.000	0.667
L15	Random forest	0.500	0.500	1.000	0.667
L5	SVM	0.500	0.000	0.000	0.000
L5	KNN	0.500	0.500	1.000	0.667
L5	XGBoost	0.500	0.500	1.000	0.667
L5	Logistic regression	0.500	0.500	1.000	0.667
L5	Random forest	0.500	0.500	1.000	0.667

Algorithm 2 has not been generated for this process on this dataset.

Early split and undersample, using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	0.500	0.000	0.000	0.000
L15	KNN	0.500	0.500	1.000	0.667
L15	XGBoost	0.500	0.500	1.000	0.667
L15	Logistic regression	0.500	0.500	1.000	0.667
L15	Random forest	0.500	0.500	1.000	0.667
L5	SVM	0.499	0.000	0.000	0.000
L5	KNN	0.500	0.500	1.000	0.667
L5	XGBoost	0.500	0.500	1.000	0.667
L5	Logistic regression	0.500	0.500	1.000	0.667
L5	Random forest	0.500	0.500	1.000	0.667

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	0.500	0.500	1.000	0.667
L15	KNN	0.500	0.500	1.000	0.667
L15	XGBoost	0.500	0.500	1.000	0.667
L15	Logistic regression	0.500	0.500	1.000	0.667
L15	Random forest	0.500	0.500	1.000	0.667
L5	SVM	0.499	0.000	0.000	0.000
L5	KNN	0.500	0.500	1.000	0.667
L5	XGBoost	0.500	0.500	1.000	0.667
L5	Logistic regression	0.500	0.500	1.000	0.667
L5	Random forest	0.500	0.500	1.000	0.667

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small p -values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L5	(2, 2)	0.0418	0.005	100	100
L5	(1, 1)	0.3139	0.005	100	100
L5	(2, 1)	0.1286	0.005	100	100
L5	(2, 2)	0.1018	0.005	100	100
L5	(1, 1)	0.6790	0.005	100	100
L5	(2, 1)	0.2380	0.005	100	100
L15	(2, 2)	0.0298	0.005	100	100
L15	(1, 1)	0.2793	0.005	100	100
L15	(2, 1)	0.1249	0.005	100	100
L15	(2, 2)	0.0102	0.005	100	100
L15	(1, 1)	0.2587	0.005	100	100
L15	(2, 1)	0.0876	0.005	100	100

Sampling-ratio audit ($H0$ and $H1$ processes only). Reuse ratio = (points per snapshot $\times$ number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
5.0	historical l500	265	500	24.98	class1
5.0	revised ceil n over t	265	21	1.05	class1
5.0	historical l500	265	500	24.98	class2

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
5.0	revised ceil n over t	265	21	1.05	class2
15.0	historical l500	795	500	74.94	class1
15.0	revised ceil n over t	795	7	1.05	class1
15.0	historical l500	795	500	74.94	class2
15.0	revised ceil n over t	795	7	1.05	class2
5.0	historical l500	66	500	24.89	class1
5.0	revised ceil n over t	66	21	1.05	class1
5.0	historical l500	66	500	24.89	class2
5.0	revised ceil n over t	66	21	1.05	class2
15.0	historical l500	198	500	74.66	class1
15.0	revised ceil n over t	198	7	1.05	class1
15.0	historical l500	198	500	74.66	class2
15.0	revised ceil n over t	198	7	1.05	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). H0 and H1 processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.733	0.733	0.706	0.800	0.750	0.782
Random forest	0.700	0.700	0.714	0.667	0.690	0.769
XGBoost	0.733	0.733	0.769	0.667	0.714	0.796
SVM	0.767	0.767	0.722	0.867	0.788	0.738
KNN	0.733	0.733	0.733	0.733	0.733	0.747
Logistic regression	0.900	0.900	0.875	0.933	0.903	0.942
Random forest	0.833	0.833	0.917	0.733	0.815	0.931
XGBoost	0.767	0.767	0.900	0.600	0.720	0.876
SVM	0.900	0.900	0.875	0.933	0.903	0.951
KNN	0.867	0.867	0.824	0.933	0.875	0.920
Logistic regression	0.967	0.967	1.000	0.933	0.966	1.000
Random forest	0.867	0.867	0.923	0.800	0.857	0.960
XGBoost	0.867	0.867	0.923	0.800	0.857	0.907
SVM	0.933	0.933	1.000	0.867	0.929	0.982

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
KNN	0.900	0.900	0.929	0.867	0.897	0.942

Early split, no undersample, using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	0.137	0.215	0.274	0.241
L15	KNN	0.500	0.500	1.000	0.667
L15	XGBoost	0.911	0.849	1.000	0.918
L15	Logistic regression	0.500	0.500	1.000	0.667
L15	Random forest	0.500	0.500	1.000	0.667
L5	SVM	0.171	0.255	0.342	0.292
L5	KNN	0.500	0.500	1.000	0.667
L5	XGBoost	0.553	0.528	1.000	0.691
L5	Logistic regression	0.500	0.500	1.000	0.667
L5	Random forest	0.500	0.500	1.000	0.667

Algorithm 2 has not been generated for this process on this dataset.

Early split, no undersample, using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	0.500	0.500	1.000	0.667
L15	KNN	0.500	0.500	1.000	0.667
L15	XGBoost	0.911	0.849	1.000	0.918
L15	Logistic regression	0.500	0.500	1.000	0.667
L15	Random forest	0.500	0.500	1.000	0.667
L5	SVM	0.500	0.500	1.000	0.667
L5	KNN	0.500	0.500	1.000	0.667
L5	XGBoost	0.553	0.528	1.000	0.691

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L5	Logistic regression	0.500	0.500	1.000	0.667
L5	Random forest	0.500	0.500	1.000	0.667

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	0.500	0.500	1.000	0.667
L15	KNN	0.500	0.500	1.000	0.667
L15	XGBoost	0.911	0.849	1.000	0.918
L15	Logistic regression	0.500	0.500	1.000	0.667
L15	Random forest	0.500	0.500	1.000	0.667
L5	SVM	0.500	0.500	1.000	0.667
L5	KNN	0.500	0.500	1.000	0.667
L5	XGBoost	0.553	0.528	1.000	0.691
L5	Logistic regression	0.500	0.500	1.000	0.667
L5	Random forest	0.500	0.500	1.000	0.667

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small *p*-values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L5	(2, 2)	0.0208	0.005	100	100
L5	(1, 1)	0.2587	0.005	100	100
L5	(2, 1)	0.1085	0.005	100	100
L5	(2, 2)	0.0666	0.005	100	100
L5	(1, 1)	0.5003	0.005	100	100
L5	(2, 1)	0.1762	0.005	100	100
L15	(2, 2)	0.0122	0.005	100	100
L15	(1, 1)	0.1900	0.005	100	100
L15	(2, 1)	0.0873	0.005	100	100
L15	(2, 2)	0.0122	0.005	100	100
L15	(1, 1)	0.2420	0.005	100	100
L15	(2, 1)	0.0891	0.005	100	100

Sampling-ratio audit (H0 and H1 processes only). Reuse ratio = (points per snapshot × number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
5.0	historical l500	265	500	24.98	class1
5.0	revised ceil n over t	265	21	1.05	class1
5.0	historical l500	933	500	24.99	class2
5.0	revised ceil n over t	933	21	1.05	class2
15.0	historical l500	795	500	74.94	class1
15.0	revised ceil n over t	795	7	1.05	class1
15.0	historical l500	2800	500	74.99	class2
15.0	revised ceil n over t	2800	7	1.05	class2
5.0	historical l500	66	500	24.89	class1
5.0	revised ceil n over t	66	21	1.05	class1
5.0	historical l500	233	500	24.96	class2
5.0	revised ceil n over t	233	21	1.05	class2
15.0	historical l500	198	500	74.66	class1
15.0	revised ceil n over t	198	7	1.05	class1
15.0	historical l500	700	500	74.99	class2
15.0	revised ceil n over t	700	7	1.05	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). H0 and H1 processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.800	0.800	0.800	0.800	0.800	0.867
Random forest	0.833	0.833	0.857	0.800	0.828	0.887
XGBoost	0.733	0.733	0.733	0.733	0.733	0.849
SVM	0.833	0.833	0.857	0.800	0.828	0.893
KNN	0.733	0.733	0.733	0.733	0.733	0.822
Logistic regression	0.867	0.867	0.923	0.800	0.857	0.947
Random forest	0.867	0.867	0.923	0.800	0.857	0.973
XGBoost	0.833	0.833	0.857	0.800	0.828	0.942
SVM	0.900	0.900	0.929	0.867	0.897	0.960
KNN	0.800	0.800	0.846	0.733	0.786	0.931
Logistic regression	0.933	0.933	0.882	1.000	0.938	0.991

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Random forest	0.933	0.933	0.882	1.000	0.938	0.980
XGBoost	0.900	0.900	0.875	0.933	0.903	0.964
SVM	0.933	0.933	0.882	1.000	0.938	1.000
KNN	0.900	0.900	0.833	1.000	0.909	0.976

Late split and undersample (the original historical run), using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	1.000	1.000	1.000	1.000
L15	KNN	1.000	1.000	1.000	1.000
L15	XGBoost	1.000	1.000	1.000	1.000
L15	Logistic regression	1.000	1.000	1.000	1.000
L15	Random forest	1.000	1.000	1.000	1.000
L5	SVM	0.975	0.990	0.960	0.975
L5	KNN	0.980	0.990	0.970	0.980
L5	XGBoost	0.980	1.000	0.960	0.980
L5	Logistic regression	0.980	0.990	0.970	0.980
L5	Random forest	0.990	1.000	0.980	0.990

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L5	SVM	0.975	0.012
L5	KNN	0.984	0.015
L5	XGBoost	0.988	0.017
L5	Logistic regression	0.974	0.014
L5	Random forest	0.988	0.015
L15	SVM	0.990	0.012
L15	KNN	0.999	0.004
L15	XGBoost	0.999	0.004

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L15	Logistic regression	0.990	0.012
L15	Random forest	1.000	0.000

Algorithm 2 has not been generated for this process on this dataset.

Late split and undersample (the original historical run), using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	1.000	1.000	1.000	1.000
L15	KNN	1.000	1.000	1.000	1.000
L15	XGBoost	1.000	1.000	1.000	1.000
L15	Logistic regression	1.000	1.000	1.000	1.000
L15	Random forest	1.000	1.000	1.000	1.000
L5	SVM	0.985	1.000	0.970	0.985
L5	KNN	0.990	1.000	0.980	0.990
L5	XGBoost	0.990	1.000	0.980	0.990
L5	Logistic regression	0.985	1.000	0.970	0.985
L5	Random forest	0.985	1.000	0.970	0.985

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L5	SVM	0.954	0.023
L5	KNN	0.985	0.015
L5	XGBoost	0.994	0.008
L5	Logistic regression	0.934	0.026
L5	Random forest	0.991	0.013
L15	SVM	0.980	0.022
L15	KNN	0.993	0.008
L15	XGBoost	0.999	0.004
L15	Logistic regression	0.974	0.025

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L15	Random forest	1.000	0.000

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	1.000	1.000	1.000	1.000
L15	KNN	1.000	1.000	1.000	1.000
L15	XGBoost	1.000	1.000	1.000	1.000
L15	Logistic regression	1.000	1.000	1.000	1.000
L15	Random forest	1.000	1.000	1.000	1.000
L5	SVM	0.985	1.000	0.970	0.985
L5	KNN	0.985	1.000	0.970	0.985
L5	XGBoost	0.990	1.000	0.980	0.990
L5	Logistic regression	0.990	1.000	0.980	0.990
L5	Random forest	0.980	1.000	0.960	0.980

Cross-validation (training-set resampling; not the held-out test score).

Snapshot / setting	Model	Mean CV accuracy	Std. deviation
L5	SVM	0.886	0.033
L5	KNN	0.989	0.013
L5	XGBoost	0.991	0.011
L5	Logistic regression	0.975	0.017
L5	Random forest	0.990	0.012
L15	SVM	0.970	0.025
L15	KNN	0.996	0.006
L15	XGBoost	0.999	0.004
L15	Logistic regression	0.974	0.025
L15	Random forest	1.000	0.000

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small p -values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L5	(2, 2)	0.0109	0.005	100	100
L5	(1, 1)	0.2200	0.005	100	100

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L5	(2, 1)	0.0822	0.005	100	100
L15	(2, 2)	0.0076	0.005	100	100
L15	(1, 1)	0.1692	0.005	100	100
L15	(2, 1)	0.0704	0.005	100	100

Sampling-ratio audit (H0 and H1 processes only). Reuse ratio = (points per snapshot × number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
5.0	historical l500	331	500	24.96	class1
5.0	revised ceil n over t	331	21	1.05	class1
5.0	historical l500	331	500	24.96	class2
5.0	revised ceil n over t	331	21	1.05	class2
15.0	historical l500	994	500	74.96	class1
15.0	revised ceil n over t	994	7	1.05	class1
15.0	historical l500	994	500	74.96	class2
15.0	revised ceil n over t	994	7	1.05	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). H0 and H1 processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.733	0.733	0.733	0.733	0.733	0.778
Random forest	0.633	0.633	0.667	0.533	0.593	0.727
XGBoost	0.500	0.500	0.500	0.467	0.483	0.613
SVM	0.700	0.700	0.714	0.667	0.690	0.751
KNN	0.667	0.667	0.647	0.733	0.688	0.778
Logistic regression	0.867	0.867	0.824	0.933	0.875	0.924
Random forest	0.867	0.867	0.789	1.000	0.882	0.960
XGBoost	0.800	0.800	0.714	1.000	0.833	0.876
SVM	0.867	0.867	0.824	0.933	0.875	0.942
KNN	0.767	0.767	0.722	0.867	0.788	0.876
Logistic regression	0.933	0.933	0.933	0.933	0.933	0.960
Random forest	0.933	0.933	0.933	0.933	0.933	0.967
XGBoost	0.900	0.900	0.929	0.867	0.897	0.951

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
SVM	0.900	0.900	0.875	0.933	0.903	0.960
KNN	0.800	0.800	0.737	0.933	0.824	0.898

Late split, no undersample, using just H0

Default classifiers on H0 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	1.000	1.000	1.000	1.000
L15	KNN	1.000	1.000	1.000	1.000
L15	XGBoost	1.000	1.000	1.000	1.000
L15	Logistic regression	1.000	1.000	1.000	1.000
L15	Random forest	1.000	1.000	1.000	1.000
L5	SVM	1.000	1.000	1.000	1.000
L5	KNN	1.000	1.000	1.000	1.000
L5	XGBoost	1.000	1.000	1.000	1.000
L5	Logistic regression	1.000	1.000	1.000	1.000
L5	Random forest	1.000	1.000	1.000	1.000

Algorithm 2 has not been generated for this process on this dataset.

Late split, no undersample, using both H0 and H1

Default classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	1.000	1.000	1.000	1.000
L15	KNN	1.000	1.000	1.000	1.000
L15	XGBoost	1.000	1.000	1.000	1.000
L15	Logistic regression	1.000	1.000	1.000	1.000
L15	Random forest	1.000	1.000	1.000	1.000
L5	SVM	1.000	1.000	1.000	1.000
L5	KNN	1.000	1.000	1.000	1.000

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L5	XGBoost	1.000	1.000	1.000	1.000
L5	Logistic regression	1.000	1.000	1.000	1.000
L5	Random forest	1.000	1.000	1.000	1.000

Retuned classifiers on H0 and H1 barcode statistics.

Snapshot / setting	Model	Accuracy	Precision	Recall	F1
L15	SVM	1.000	1.000	1.000	1.000
L15	KNN	1.000	1.000	1.000	1.000
L15	XGBoost	1.000	1.000	1.000	1.000
L15	Logistic regression	1.000	1.000	1.000	1.000
L15	Random forest	1.000	1.000	1.000	1.000
L5	SVM	1.000	1.000	1.000	1.000
L5	KNN	1.000	1.000	1.000	1.000
L5	XGBoost	1.000	1.000	1.000	1.000
L5	Logistic regression	1.000	1.000	1.000	1.000
L5	Random forest	1.000	1.000	1.000	1.000

Robinson–Turner Algorithm 2 permutation test on barcode-vector proxies. Small p -values mean the class contrast is unusual under a random label shuffle.

Snapshot	Contrast (p, q)	Observed statistic	p-value	n1	n2
L5	(2, 2)	0.0180	0.005	100	100
L5	(1, 1)	0.2157	0.005	100	100
L5	(2, 1)	0.0933	0.005	100	100
L15	(2, 2)	0.0081	0.005	100	100
L15	(1, 1)	0.1584	0.005	100	100
L15	(2, 1)	0.0741	0.005	100	100

Sampling-ratio audit (H0 and H1 processes only). Reuse ratio = (points per snapshot $\times$ number of snapshots) / minority class count.

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
5.0	historical l500	331	500	24.96	class1
5.0	revised ceil n over t	331	21	1.05	class1
5.0	historical l500	1166	500	24.98	class2
5.0	revised ceil n over t	1166	21	1.05	class2

Snapshot size %	Rule	Points / snapshot	Number of snapshots	Reuse ratio	Class
15.0	historical l500	994	500	74.96	class1
15.0	revised ceil n over t	994	7	1.05	class1
15.0	historical l500	3500	500	74.99	class2
15.0	revised ceil n over t	3500	7	1.05	class2

Revised snapshot protocol (fixed points per snapshot; default 60 training / 15 test snapshots). H0 and H1 processes only.

Model	Accuracy	Balanced acc.	Precision	Recall	F1	ROC AUC
Logistic regression	0.833	0.833	0.812	0.867	0.839	0.867
Random forest	0.767	0.767	0.750	0.800	0.774	0.849
XGBoost	0.667	0.667	0.647	0.733	0.688	0.764
SVM	0.833	0.833	0.857	0.800	0.828	0.884
KNN	0.733	0.733	0.706	0.800	0.750	0.804
Logistic regression	0.833	0.833	0.857	0.800	0.828	0.938
Random forest	0.867	0.867	0.867	0.867	0.867	0.964
XGBoost	0.867	0.867	0.867	0.867	0.867	0.956
SVM	0.867	0.867	0.923	0.800	0.857	0.929
KNN	0.800	0.800	0.800	0.800	0.800	0.911
Logistic regression	1.000	1.000	1.000	1.000	1.000	1.000
Random forest	0.900	0.900	0.875	0.933	0.903	0.982
XGBoost	0.933	0.933	0.933	0.933	0.933	0.987
SVM	0.967	0.967	0.938	1.000	0.968	0.996
KNN	0.900	0.900	0.929	0.867	0.897	0.964